

expected returns. We periodically test this proposition, so far without much success.

Aswath Damodaran, professor of finance at the Stern School of Business at New York University, has found that the enterprise multiple, along with the price-to-earnings ratio and price-to-sales ratio, is the most common relative valuation ratio used by equity analysts.¹⁹ Driven by the popularity of the enterprise multiple in industry, other researchers have challenged Fama and French's positions that investors can remain agnostic or that price-to-book value is the best metric. In 2009, Tim Loughran, the C.R. Smith professor of finance at the University of Notre Dame, and Jay W. Wellman, the visiting assistant professor of finance at Cornell University, highlighted serious flaws with the use of price-to-book value, arguing that the enterprise multiple was the better metric for several important reasons.²⁰ The primary reason is that the enterprise multiple better identified undervalued stocks over the full period of Loughran and Wellman's research—1963 to 2008—and better identified undervalued large market capitalization stocks where they found price-to-book value to be incompetent. Note that Loughran and Wellman define the enterprise multiple in a slightly different way to Greenblatt, employing earnings before interest, taxes, depreciation, and amortization (EBITDA) in place of Greenblatt's EBIT:

$$\text{Enterprise Multiple} = \text{EBITDA} \div \text{EV}$$

As we'll see, whether we employ EBIT or EBITDA makes little difference. The major problem with price-to-book value is that it tends to identify very small, micro-capitalization stocks and stocks that benefit from the so-called "January effect"—a general increase in stock prices in January, possibly as a result of tax-loss selling in December. Loughran and Wellman found that for the largest stocks, which account for about 94 percent of the stock market's total market capitalization, price-to-book value could not be used to identify undervalued stocks once they controlled for the January effect. Thus, conclude Loughran and Wellman, for nearly the entire market value in the world's largest stock market (the United States), over the most important time period (post-1963), price-to-book value could not identify undervalued stocks. Additionally, price-to-book value may not have identified undervalued stocks in the smallest 6 percent of stocks because the wide bid-ask spreads may have misrepresented the price at which those micro-capitalization stocks could be traded. Loughran and Wellman also note that many investment managers are restricted in their ability to buy very small stocks due to ownership concentration restrictions, and are prohibited from buying low-priced stocks—a stock priced below \$5—due to their